

## Exercises 10.B

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**Exercise 1.** Suppose  $V$  is a real vector space. Suppose  $T \in \mathcal{L}(V)$  has no eigenvalues. Prove that  $\det T > 0$ .

*Proof.* Since  $T$  has no eigenvalues,  $T_C$  has no "real" eigenvalues. That is, each eigenvalue of  $T_C$  has a nontrivial real part. This implies that all the eigenvalues of  $T_C$  come in pairs of a complex number and its conjugate, i.e. if  $\lambda$  is an eigenvalue of  $T_C$ , then  $\bar{\lambda}$  is also an eigenvalue of  $T_C$ . Then the determinant of  $T$ , which is the product of the eigenvalues of  $T_C$ , is of the form

$$(a_1 + b_1 i)(a_1 - b_1 i) \cdots (a_n + b_n i)(a_n - b_n i),$$

where  $a_j, b_j \in \mathbb{R}$  for all  $j \in \{1, \dots, n\}$ . Simplifying this, we get

$$(a_1^2 + b_1^2) \cdots (a_n^2 + b_n^2),$$

which is a product of *positive* integers since  $a_j$  and  $b_j$  are not both 0 for a given  $j \in \{1, \dots, n\}$  (otherwise 0 would be an eigenvalue of  $T$ ). Thus the product, which is the determinant, is positive.  $\square$

**Exercise 2.** Suppose  $V$  is a real vector space with even dimension and  $T \in \mathcal{L}(V)$ . Suppose  $\det T < 0$ . Prove that  $T$  has at least two distinct eigenvalues.

*Proof.* From Exercise 1, we know that  $T$  has at least one eigenvalue since if it had none the determinant would be positive. Since complex eigenvalues of  $T_C$  come in pairs, the sum of the multiplicities of the eigenvalues unique to  $T_C$  will be even. Since the sum of the multiplicities of all the eigenvalues of  $T_C$  is  $\dim V$ , which is even, this means that the sum of the multiplicities which are shared between  $T_C$  and  $T$ , namely the real eigenvalues, is even. Note that 0 is not an eigenvalue of  $T$ . Also, by the proof of Exercise 1, we know that the product of the "complex" eigenvalues of  $T_C$  is positive, so the product of the real eigenvalues must be negative.

Assume that  $\lambda$ , our one known eigenvalue of  $T$ , is in fact the only eigenvalue of  $T$ . Then  $n = \dim G(\lambda, T_C)$  is even as noted above, which would mean that  $\det T = \lambda^n p$  where  $p$  is the product of the complex eigenvalues of  $T_C$  (repeated according to their multiplicity), and is thus positive. Of course, since  $\lambda$  is real,  $\lambda^n$  is positive if  $n$  is even, which would imply that  $\det T > 0$ , contradicting the hypothesis that  $\det T < 0$ . Thus, there are at least two distinct eigenvalues of  $T$ .  $\square$

**Exercise 3.** Suppose  $T \in \mathcal{L}(V)$  and  $n = \dim V > 2$ . Let  $\lambda_1, \dots, \lambda_n$  denote the eigenvalues of  $T$  (or of  $T_C$  if  $V$  is a real vector space), repeated according to multiplicity.

(a) Find a formula for the coefficient of  $z^{n-2}$  in the characteristic polynomial of  $T$  in terms of  $\lambda_1, \dots, \lambda_n$ .

(b) Find a formula for the coefficient of  $z$  in the characteristic polynomial of  $T$  in terms of  $\lambda_1, \dots, \lambda_n$ .

*Proof.*

(a) If  $n = 3$ , then we have

$$\begin{aligned} (z - \lambda_1)(z - \lambda_2)(z - \lambda_3) &= (z - \lambda_1)(z^2 - \lambda_2 z - \lambda_3 z + \lambda_2 \lambda_3) \\ &= z^3 - \lambda_2 z^2 - \lambda_3 z^2 + \lambda_2 \lambda_3 z - \lambda_1 z^2 + \\ &\quad \lambda_1 \lambda_2 z + \lambda_1 \lambda_3 z - \lambda_1 \lambda_2 \lambda_3 \\ &= z^3 - (\lambda_1 + \lambda_2 + \lambda_3)z^2 + \\ &\quad (\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_1 \lambda_3)z - \lambda_1 \lambda_2 \lambda_3. \end{aligned}$$

We'll use this as a base case for induction in the proofs of both (a) and (b). Define  $p_m = (z - \lambda_1) \cdots (z - \lambda_m)$  for  $m \in \{1, \dots, n\}$ , and define  $c_{m,\ell}$  to be the coefficient of  $z^\ell$  in  $p_m$ . We also define  $C_m = \{\{j, k\} \mid j, k \in \{1, \dots, m\}, j \neq k\}$ . We propose that the coefficient of  $z^{n-2}$  is of the form

$$\sum_{\{j,k\} \in C_n} \lambda_j \lambda_k.$$

To prove this, first observe from the above proposed base case that the hypothesis does in fact hold for  $n = 3$ , so we do in fact have a base case. Now assume that  $n > 3$  and the hypothesis holds for all values less than  $n$ . Then we can write

$$p_n = (z - \lambda_n)p_{n-1}, \quad (1)$$

so long as each  $\lambda_j$  is indexed appropriately. Observe by (1) that  $c_{n,n-2}$  will be precisely  $c_{n-1,n-3} - \lambda_n c_{n-1,n-2}$ . We know that  $c_{n-1,n-2} = -\sum_{j=1}^{n-1} \lambda_j$ , and we also know that the hypothesis holds for  $c_{n-1,n-3}$ . Thus

$$c_{n,n-2} = \left( \sum_{\{j,k\} \in C_{n-1}} \lambda_j \lambda_k \right) + \left( \sum_{\ell=1}^{n-1} \lambda_n \lambda_\ell \right). \quad (2)$$

Since the first summand of (2) contains all necessary combinations of  $\lambda_1, \dots, \lambda_{n-1}$  and the second contains all combinations of  $\lambda_n$  with  $\lambda_1, \dots, \lambda_{n-1}$ , we can see that in fact (2) is

$$\sum_{\{j,k\} \in C_n} \lambda_j \lambda_k$$

as desired.

(b) If  $n = 4$ , we can take this and get

$$\begin{aligned}
(z - \lambda_1)(z - \lambda_2)(z - \lambda_3)(z - \lambda_4) &= (z - \lambda_4)(z^3 - (\lambda_1 + \lambda_2 + \lambda_3)z^2 + \\
&\quad (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3)z - \lambda_1\lambda_2\lambda_3) \\
&= z^4 - (\lambda_1 + \lambda_2 + \lambda_3)z^3 + \\
&\quad (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3)z^2 - \lambda_1\lambda_2\lambda_3z - \\
&\quad \lambda_4z^3 + (\lambda_1\lambda_4 + \lambda_2\lambda_4 + \lambda_3\lambda_4)z^2 - \\
&\quad (\lambda_1\lambda_2\lambda_4 + \lambda_2\lambda_3\lambda_4 + \lambda_1\lambda_3\lambda_4)z + \\
&\quad \lambda_1\lambda_2\lambda_3\lambda_4 \\
&= z^4 - (\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)z^3 + \\
&\quad (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3 + \\
&\quad \lambda_1\lambda_4 + \lambda_2\lambda_4 + \lambda_3\lambda_4)z^2 - \\
&\quad (\lambda_1\lambda_2\lambda_4 + \lambda_2\lambda_3\lambda_4 + \lambda_1\lambda_3\lambda_4)z + \\
&\quad \lambda_1\lambda_2\lambda_3\lambda_4
\end{aligned}$$

Do a similar god-forsaken combinatorial procedure as that in the proof of (a) and you will see (probably) that  $c_{n,1}$  is the sum of all three-item combinations of  $\lambda_1, \dots, \lambda_n$ .  $\square$

**Exercise 4.** Suppose  $T \in \mathcal{L}(V)$  and  $c \in \mathbf{F}$ . Prove that  $\det(cT) = c^{\dim V} \det T$ .

*Proof.* Since  $\det T = \det \mathcal{M}(T)$ , it will suffice to consider the determinant of  $\mathcal{M}(cT) = c\mathcal{M}(T)$ . Note that  $c\mathcal{M}(T)$  multiplies every column of  $\mathcal{M}(T)$  by  $c$ . Applying 10.39 successively to each column reveals that

$$\det(c\mathcal{M}(T)) = c^{\dim V} \det \mathcal{M}(T)$$

since there are  $\dim V$  columns of  $\mathcal{M}(T)$ .  $\square$

**Exercise 5.** Prove or give a counterexample: if  $S, T \in \mathcal{L}(V)$ , then  $\det(S+T) = \det S + \det T$ .

*Proof.* Consider  $S = T = I$ . Then

$$\begin{aligned}
\det(S + T) &= \det(\mathcal{M}(S + T)) \\
&= \det(\mathcal{M}(S) + \mathcal{M}(T)) \\
&= \det\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) \\
&= \det\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\
&= 4
\end{aligned}$$

Since  $\det S = \det T = 1$ ,  $\det S + \det T = 2$ . Thus

$$\det(S + T) = 4 \neq 2 = \det S + \det T.$$

$\square$

**Exercise 6.** Suppose  $A$  is a block upper-triangular matrix

$$A = \begin{pmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where each  $A_j$  along the diagonal is a square matrix. Prove that

$$\det A = (\det A_1) \cdots (\det A_m).$$

**Exercise 7.** Suppose  $A$  is an  $n$ -by- $n$  matrix with real entries. Let  $S \in \mathcal{L}(\mathbb{C}^n)$  denote the operator on  $\mathbb{C}^n$  whose matrix equals  $A$ , and let  $T \in \mathcal{L}(\mathbb{R}^n)$  denote the operator on  $\mathbb{R}^n$  whose matrix equals  $A$ . Prove that  $\text{trace } S = \text{trace } T$  and  $\det S = \det T$ .

*Proof.* Both statements follow trivially from the fact that  $\mathcal{M}(T) = \mathcal{M}(T_C)$ .  $\square$

**Exercise 8.** Suppose  $V$  is an inner product space and  $T \in \mathcal{L}(V)$ . Prove that

$$\det T^* = \overline{\det T}.$$

Use this to prove that  $|\det T| = \det \sqrt{T^*T}$ , giving a different proof than was given in 10.47.

*Proof.* Consider a basis of  $V$  as in 8.29. Let  $n = \dim V$  and  $\lambda_1, \dots, \lambda_n$  the entries along the diagonal of the matrix of  $T$  with respect to our chosen basis, i.e. the eigenvalues of  $T$  repeated according to their multiplicity. Then  $\det T = \det \mathcal{M}(T) = \lambda_1 \cdots \lambda_n$ . Note that  $\mathcal{M}(T^*)$  is the conjugate transpose of  $\mathcal{M}(T)$ , so the diagonal of  $\mathcal{M}(T^*)$  is  $\overline{\lambda_1}, \dots, \overline{\lambda_n}$ . Since  $\mathcal{M}(T^*)$  is "lower-triangular", we can easily see that the only term which is nonzero in the definition of the determinant is the product of the entries along the diagonal. Thus

$$\det T^* = \overline{\lambda_1} \cdots \overline{\lambda_n} = \overline{\lambda_1 \cdots \lambda_n} = \overline{\det T}.$$

For the second part of the exercise, we proceed with

$$\begin{aligned} |\det T| &= \sqrt{\det T^* \det T} \\ &= \sqrt{\det T^* \det T} \\ &= \sqrt{\det T^* T}. \end{aligned} \tag{1}$$

This gets us close, but not all the way there. Now we need to show that  $\sqrt{\det T^* T} = \det \sqrt{T^* T}$ .

Redefine  $\lambda_1, \dots, \lambda_n$  to be the eigenvalues of  $T^* T$ , repeated according to multiplicity. Note that the eigenvalues of  $\sqrt{T^* T}$  (repeated according to multiplicity) are  $\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}$ . So

$$\begin{aligned} \sqrt{\det T^* T} &= \sqrt{\lambda_1 \cdots \lambda_n} \\ &= \sqrt{\lambda_1} \cdots \sqrt{\lambda_n} \\ &= \det \sqrt{T^* T} \end{aligned}$$

as desired.

□